CSE 150A-250A AI: Probabilistic Models

Lecture 9

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Agenda

Review

Incomplete Data

Expectation-Maximization Algorithm

Review

Learning in BNs with discrete nodes

- ML estimation for complete data:

$$P_{\mathrm{ML}}(X_i = x | \mathrm{pa}_i = \pi) = \frac{\mathrm{count}(X_i = x, \mathrm{pa}_i = \pi)}{\sum_{x'} \mathrm{count}(X_i = x', \mathrm{pa}_i = \pi)}$$

- For nodes with parents:

$$P_{\mathrm{ML}}(X_i = x | \mathrm{pa}_i = \pi) = \frac{\mathrm{count}(X_i = x, \mathrm{pa}_i = \pi)}{\mathrm{count}(\mathrm{pa}_i = \pi)}$$

- For root nodes:

$$P_{\mathrm{ML}}(X_i = x) = \frac{\mathrm{count}(X_i = x)}{T}$$

Markov models for statistical language processing

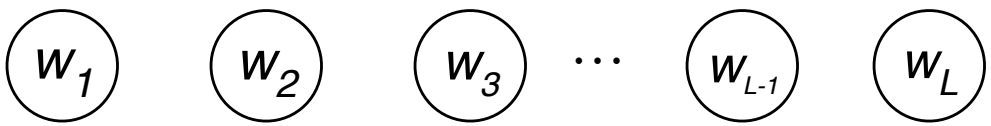
- n-gram models of word sequences:

$$P(w_1, w_2, \ldots, w_L) = \prod_{\ell} P(w_\ell | \underbrace{w_{\ell-(n-1}, \ldots, w_{\ell-1})}_{\text{previous words}})$$

- As belief networks:

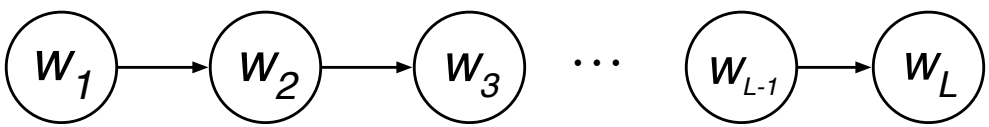
$n = 1$

unigram



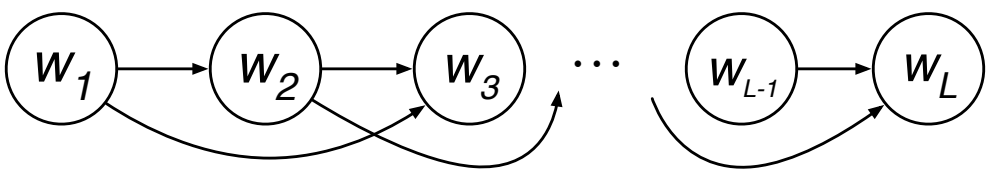
$n = 2$

bigram



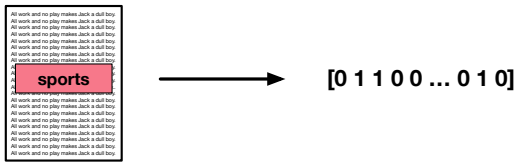
$n = 3$

trigram



Naive Bayes model for document classification

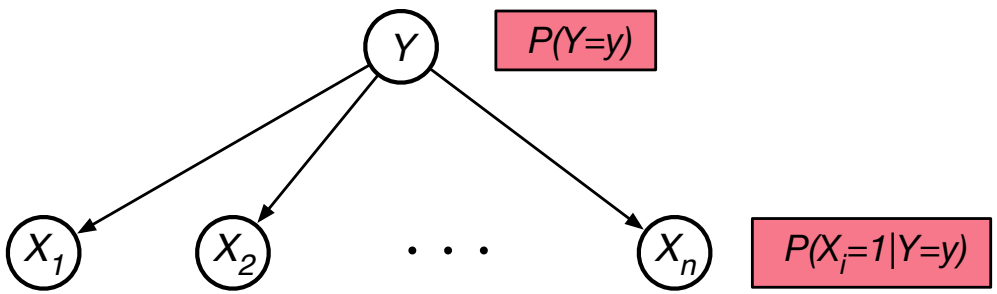
- Random variables



$Y \in \{1, 2, \ldots, m\}$ topic of document

$X_i \in \{0, 1\}$ i^{th} word appears?

- Belief network



- Naive Bayes assumption

$$P(X_1, \ldots, X_n | Y) = \prod_{i=1}^{n} P(X_i | Y)$$

Incomplete Data

Learning from incomplete data with tabular CPTs

ASSUMPTIONS

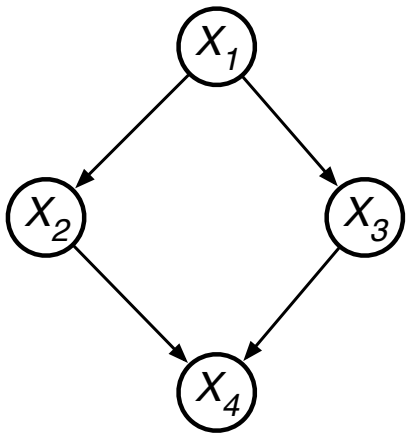
1. The DAG is fixed (and known) over a finite set of discrete random variables $\{X_1, X_2, \ldots, X_n\}$.

2. CPTs enumerate $P(X_i = x | \text{pa}(X_i) = \pi)$ as lookup tables; each must be estimated for all values of x and π.

3. The data is IID, but only consists of T partially complete instantiations of the nodes in the BN.

Toy example

- Fixed DAG over binary random variables



$$X_1 \in \{0,1\}$$
$$X_2 \in \{0,1\}$$
$$X_3 \in \{0,1\}$$
$$X_4 \in \{0,1\}$$

- Incomplete data set

example	X_1	X_2	X_3	X_4
1	1	?	0	1
2	0	1	?	0
3	?	?	?	1
:	:	:	:	:
T	?	1	1	0

How to choose the CPTs so that the BN maximizes the probability of this data set?

A more interesting example ...















How to build a movie recommendation system?

- Collect a data set of movie ratings:

$$\begin{bmatrix} + & - & + & - & ? & ? & + \\ - & ? & ? & + & + & ? & ? \\ + & + & + & + & + & + & + \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ - & - & - & - & - & ? & - \\ ? & ? & + & ? & ? & ? & - \end{bmatrix}$$

$+$	liked
$-$	disliked
?	not seen

(user-item matrix)

- Build a model of user profiles and fill in the missing ratings.
  But what model to build?

Naive Bayes model with incomplete data



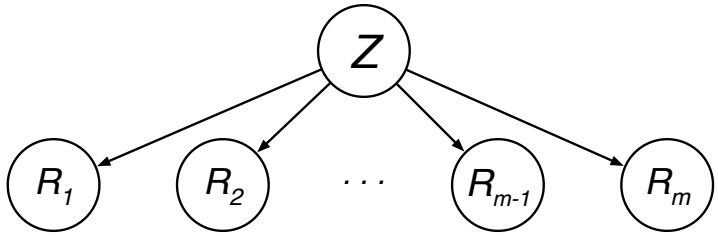










- Movie recommender system

$$
\begin{aligned}
Z &\in \{1, 2, \ldots, k\} && \text{type of movie-goer} \\
R_i &\in \{0, 1\} && \text{rating for } i^{\text{th}} \text{ movie}
\end{aligned}
$$

- Incomplete data set

student	Z	R_1	R_2	R_3	R_4	$\cdots$
1	?	0	1	1	?	$\cdots$
2	?	1	?	0	1	$\cdots$
3	?	0	0	?	1	$\cdots$
:	:	:	:	:	:	:
T	?	?	1	0	?	$\cdots$

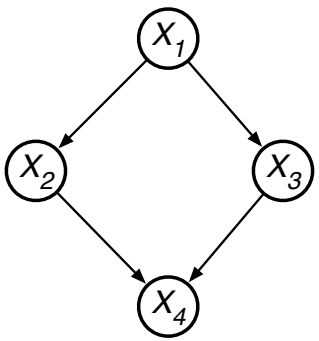
Note that the variable Z is never observed.

Learning from incomplete data

- Notation

$$H_t = \text{set of hidden (latent) variables for } t^{\text{th}} \text{ example}$$
$$V_t = \text{set of visible (observed) variables for } t^{\text{th}} \text{ example}$$

- Illustration



example	X_1	X_2	X_3	X_4
1	1	?	0	1
2	0	1	?	0
3	?	?	?	1
:	:	:	:	:

$$H_1 = \{X_2\} \qquad V_1 = \{X_1, X_3, X_4\}$$
$$H_2 = \{X_3\} \qquad V_2 = \{X_1, X_2, X_4\}$$
$$H_3 = \{X_1, X_2, X_3\} \qquad V_3 = \{X_4\}$$

Computing the log-likelihood with incomplete data

$$
\begin{aligned}
\mathcal{L} &= \log P(\text{data}) \\
&= \log \prod_{t=1}^{T} P\left(V_t = v_t\right) && \boxed{\text{data is IID}} \\
&= \sum_{t=1}^{T} \log P\left(V_t = v_t\right) && \boxed{\log ab = \log a + \log b}
\end{aligned}
$$

Q. What should we do next? (to express the full joint)

A. Use product rule

B. Express $P\left(V_t = v_t\right)$ using conditional independence

C. Use marginalization

D. Use Bayes Rule

E. None of them

Computing the log-likelihood with incomplete data

$$
\begin{aligned}
\mathcal{L} &= \log P(\text{data}) \\
&= \log \prod_{t=1}^{T} P\left(V_t = v_t\right) && \boxed{\text{data is IID}} \\
&= \sum_{t=1}^{T} \log P\left(V_t = v_t\right) && \boxed{\log ab = \log a + \log b} \\
&= \sum_{t=1}^{T} \log \sum_{h} P\left(H_t = h, V_t = v_t\right) && \boxed{\text{marginalization}} \\
&= \sum_{t=1}^{T} \log \sum_{h} P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)\Big|_{\{H_t = h, V_t = v_t\}} && \boxed{\text{joint}} \\
&= \sum_{t=1}^{T} \log \sum_{h} \prod_{i=1}^{n} P(X_i = x_i | \text{pa}_i = \pi_i)\Big|_{\{H_t = h, V_t = v_t\}} && \boxed{\text{product rule}}
\end{aligned}
$$

Computing the log-likelihood with incomplete data

$$
\begin{aligned}
\mathcal{L} &= \log P(\text{data}) \\
&= \log \prod_{t=1}^{T} P\left(V_t = v_t\right) && \boxed{\text{data is IID}} \\
&= \sum_{t=1}^{T} \log P\left(V_t = v_t\right) && \boxed{\log ab = \log a + \log b} \\
&= \sum_{t=1}^{T} \log \sum_{h} P\left(H_t = h, V_t = v_t\right) && \boxed{\text{marginalization}} \\
&= \sum_{t=1}^{T} \log \sum_{h} P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)\Big|_{\{H_t = h, V_t = v_t\}} && \boxed{\text{joint}} \\
&= \sum_{t=1}^{T} \log \sum_{h} \prod_{i=1}^{n} P(X_i = x_i | \text{pa}_i = \pi_i)\Big|_{\{H_t = h, V_t = v_t\}} && \boxed{\text{product rule}}
\end{aligned}
$$

Complete versus incomplete data

- **Complete data**

$$\mathcal{L} = \sum_{i,\pi,x} \text{count}(X_i = x, \text{pa}_i = \pi) \log P(X_i = x | \text{pa}_i = \pi)$$

The CPTs at different nodes are decoupled!
We can compute ML estimates in closed form.

- **Incomplete data**

$$\mathcal{L} = \sum_{t=1}^{T} \log \sum_{h} \prod_{i=1}^{n} P(X_i = x_i | \text{pa}_i = \pi_i) \Big|_{\{H_t = h, V_t = v_t\}}$$

The CPTs are potentially all coupled.
How to proceed?

Expectation-Maximization Algorithm

EM algorithm in a nutshell

- If only the data weren't incomplete ...

student	Z	R_1	R_2	$\cdots$
1	?	0	1	$\cdots$
2	?	1	?	$\cdots$
3	?	0	0	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
T	?	?	?	$\cdots$

If the data were complete, we could easily estimate the CPTs. What can we do instead?

- Here's a crazy idea ...

Randomly initialize the CPTs with nonzero elements.
Use these CPTs to infer values for the missing data.
Re-estimate CPTs from the newly completed data.
Iterate the last two steps until convergence?

Amazingly, this is how EM works (more or less) ...

EM algorithm — overview

- **Initialize the CPTs**

  Assign random probabilities to all $P(X_i = x | \mathrm{pa}_i = \pi)$.

  Avoid zero probabilities (which cannot be unlearned).

  Different initializations may yield different results.

- **Iterate until convergence**

  [E-Step] Compute posterior probabilities $P(H_t = h | V_t = v_t)$.

  [M-Step] Update CPTs based on these probabilities.

E-step (Inference)

To fill in missing data, we must compute posterior probabilities. But which probabilities, specifically, do we need?

$$\begin{aligned} \textbf{At root nodes:} \quad & P(X_i = x | V_t = v_t) \\ \textbf{At other nodes:} \quad & P(X_i = x, \mathrm{pa}_i = \pi | V_t = v_t) \end{aligned}$$

These probabilities must be computed over a quadruple loop:

$$\begin{aligned} \textbf{examples } V_t \quad & t \in \{1, 2, \ldots, T\} \\ \textbf{nodes } X_i \quad & i \in \{1, 2, \ldots, n\} \\ \textbf{values of } X_i = x \quad & \text{e.g., } x \in \{0, 1\} \\ \textbf{values of } \mathrm{pa}_i = \pi \quad & \text{e.g., } \pi \in \{0, 1\}^k \end{aligned}$$

The # of computations grows linearly in the size of the BN, and also in the amount of data (as expected).

M-step (Learning)

Next we use these posterior probabilities to update CPTs:

- At root nodes

$$P(X_i = x) \quad \longleftarrow \quad \frac{1}{T} \sum_{t=1}^{T} P(X_i = x | V_t = v_t)$$

- At nodes with parents

$$P(X_i = x | \mathrm{pa}_i = \pi) \quad \longleftarrow \quad \frac{\sum_{t=1} P(X_i = x, \mathrm{pa}_i = \pi | V_t = v_t)}{\sum_{t=1}^{T} P(\mathrm{pa}_i = \pi | V_t = v_t)}$$

Note that these are updates ($\longleftarrow$), not equalities ($=$).
The right hand sides depend on the current CPTs.

Formulas are great, but what about intuition?

Analogy to ML for complete data

- Indicator functions

$$I(x, x') = \begin{cases} 1 & \text{if } x = x' \\ 0 & \text{otherwise} \end{cases}$$

- Counts

$$\text{count}(X_i = x) = \sum_{t=1}^{T} I(x_{it}, x)$$

$$\text{count}(\text{pa}_i = \pi) = \sum_{t=1}^{T} I(\text{pa}_{it}, \pi)$$

$$\text{count}(X_i = x, \text{pa}_i = \pi) = \sum_{t=1}^{T} I(x_{it}, x)\, I(\text{pa}_{it}, \pi)$$

ML estimates for complete data

- At root nodes

$$P_{\mathrm{ML}}(X_i = x) = \frac{\mathrm{count}(X_i = x)}{T}$$

$$P_{\mathrm{ML}}(X_i = x) = \frac{1}{T} \sum_{t=1}^{T} I(x_{it}, x)$$

- At nodes with parents

$$P_{\mathrm{ML}}(X_i = x | \mathrm{pa}_i = \pi) = \frac{\mathrm{count}(X_i = x, \mathrm{pa}_i = \pi)}{\mathrm{count}(\mathrm{pa}_i = \pi)}$$

$$P_{\mathrm{ML}}(X_i = x | \mathrm{pa}_i = \pi) = \frac{\sum_{t=1}^{T} I(x_{it}, x)\, I(\mathrm{pa}_{it}, \pi)}{\sum_{t=1}^{T} I(\mathrm{pa}_{it}, \pi)}$$

Intuition for EM updates — by analogy

- At root nodes

$$P_{\mathrm{ML}}(X_i = x) \quad = \quad \frac{1}{T} \sum_t I(x_{it}, x)$$

ML for complete data

$$P(X_i = x) \quad \leftarrow \quad \frac{1}{T} \sum_t P(X_i = x | V_t = v_t)$$

EM update

- At nodes with parents

$$P_{\mathrm{ML}}(X_i = x | \mathrm{pa}_i = \pi) \quad = \quad \frac{\sum_t I(x_{it}, x)\, I(\mathrm{pa}_{it}, \pi)}{\sum_t I(\mathrm{pa}_{it}, \pi)}$$

ML for complete data

$$P(X_i = x | \mathrm{pa}_i = \pi) \quad \leftarrow \quad \frac{\sum_t P(X_i = x, \mathrm{pa}_i = \pi | V_t = v_t)}{\sum_t P(\mathrm{pa}_i = \pi | V_t = v_t)}$$

EM update

- Special case

Consider a CPT whose nodes are fully observed.
EM updates in this case reduce to ML estimates for complete data.

EM updates

$$P(X_i = x) \longleftarrow \frac{1}{T}\sum_t P(X_i = x | V_t = v_t) \quad \text{root nodes}$$

$$P(X_i = x | \mathrm{pa}_i = \pi) \longleftarrow \frac{\sum_t P(X_i = x, \mathrm{pa}_i = \pi | V_t = v_t)}{\sum_t P(\mathrm{pa}_i = \pi | V_t = v_t)} \quad \text{nodes with parents}$$

Intuitively:

When the data is complete, we estimate the CPTs from observed counts.

When the data is incomplete, we re-estimate the CPTs from expected counts.

These expected counts are computed from the posterior distributions $P(h|v_t)$.

Key properties of EM

- **No learning rate**

  The updates do not require the tuning of a learning rate ($\eta > 0$), as in most gradient-based methods.

- **Monotonic convergence**

  The updated CPTs from EM always increase the incomplete-data log-likelihood $\mathcal{L} = \sum_t \log P(V_t = v_t)$.

Check In

Q. How much of EM did you understand?

A. (Nearly) All of it

B. Some of it, but I have some doubts

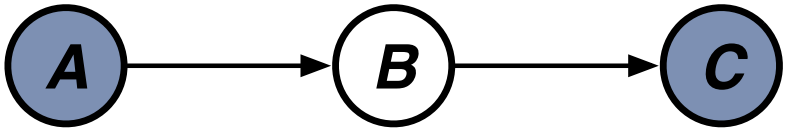
C. Maybe a little, but I'm pretty confused

D. Almost none of it; I'm totally lost

Log-likelihood

- Incomplete data set

t	A	B	C
1	a_1	?	c_1
2	a_2	?	c_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$
T	a_T	?	c_T



How to choose the CPTs
to maximize the log-likelihood
of this (incomplete) data?

- Log-likelihood

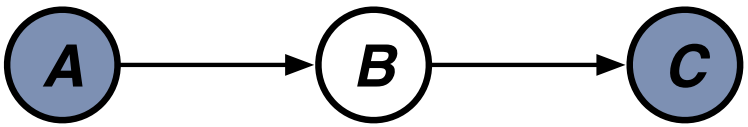
$$\mathcal{L} = \sum_t \log P(a_t, c_t)$$

$$= \sum_t \log \sum_b P(a_t, b, c_t) \quad \boxed{\text{marginalization}}$$

$$= \sum_t \log \sum_b P(a_t)\, P(b|a_t)\, P(c_t|a_t, b) \quad \boxed{\text{product rule}}$$

$$= \sum_t \log \sum_b P(a_t)\, P(b|a_t)\, P(c_t|b) \quad \boxed{\text{conditional independence}}$$

Example



Suppose that A and C are observed and B is hidden.

Q. Which parameters of this network can you estimate directly from the data (in one step—no iteration required)?

A. $P(A)$

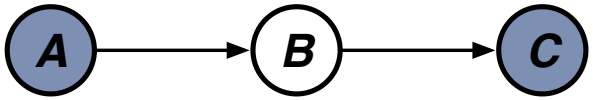
B. $P(B|A)$

C. $P(C|B)$

D. Both A. and C.

E. None of them

EM update for $P(A)$



- General form

$$P(X_i = x) \longleftarrow \frac{1}{T}\sum_t P(X_i = x | V_t = v_t) \qquad \boxed{\text{root node}}$$

- Update for this CPT

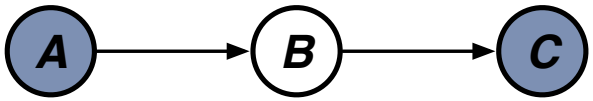
$$P(A = a) \longleftarrow \frac{1}{T}\sum_t P(A = a | A = a_t, C = c_t)$$

Simplify:

$$P(A = a) \longleftarrow \frac{1}{T}\sum_t I(a, a_t) = \frac{1}{T}\,\mathrm{count}(A = a)$$

The update reduces to the ML estimate for complete data—as it must, because A is observed and has no unobserved parents.

EM update for $P(B|A)$



- General form

$$P(X_i = x | \mathrm{pa}_i = \pi) \longleftarrow \frac{\sum_t P(X_i = x, \mathrm{pa}_i = \pi | V_t = v_t)}{\sum_t P(\mathrm{pa}_i = \pi | V_t = v_t)}$$

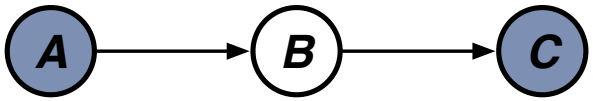
- Update for this CPT

$$P(B = b | A = a) \longleftarrow \frac{\sum_t P(B = b, A = a | A = a_t, C = c_t)}{\sum_t P(A = a | A = a_t, C = c_t)}$$

Simplify:

$$P(B = b | A = a) \longleftarrow \frac{\sum_t I(a, a_t) P(B = b | A = a_t, C = c_t)}{\sum_t I(a, a_t)}$$

EM update for $P(B|A)$



- General form

$$P(X_i = x | \mathrm{pa}_i = \pi) \longleftarrow \frac{\sum_t P(X_i = x, \mathrm{pa}_i = \pi | V_t = v_t)}{\sum_t P(\mathrm{pa}_i = \pi | V_t = v_t)}$$

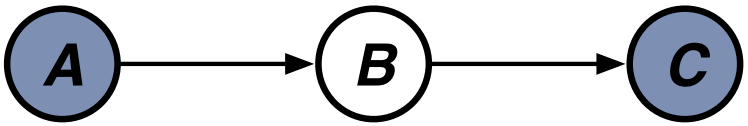
- Update for this CPT

$$P(B = b | A = a) \longleftarrow \frac{\sum_t P(B = b, A = a | A = a_t, C = c_t)}{\sum_t P(A = a | A = a_t, C = c_t)}$$

Simplify:

$$P(B = b | A = a) \longleftarrow \frac{\sum_t I(a, a_t) \overbrace{P(B = b | A = a_t, C = c_t)}^{\text{computed from Bayes rule}}}{\sum_t I(a, a_t)}$$

Example



Suppose that A and C are observed and B is hidden.

- **Inference**

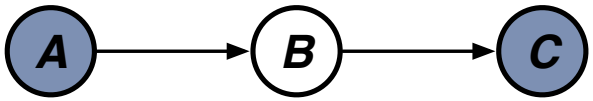
$$P(B=b|A=a,C=c) = \frac{P(C=c|B=b,A=a)\,P(B=b|A=a)}{P(C=c|A=a)} \quad \boxed{\text{BR}}$$

$$= \frac{P(C=c|B=b)\,P(B=b|A=a)}{P(C=c|A=a)} \quad \boxed{\text{CI}}$$

$$= \frac{P(C=c|B=b)\,P(B=b|A=a)}{\sum_{b'} P(C=c|B=b')\,P(B=b'|A=a)} \quad \boxed{\text{normalized}}$$

This is the only non-trivial posterior probability that we'll need for the EM updates in this example.

EM update for $P(C|B)$



- General form

$$P(X_i = x | \mathrm{pa}_i = \pi) \longleftarrow \frac{\sum_t P(X_i = x, \mathrm{pa}_i = \pi | V_t = v_t)}{\sum_t P(\mathrm{pa}_i = \pi | V_t = v_t)}$$

- Update for this CPT

$$P(C = c | B = b) \longleftarrow \frac{\sum_t P(C = c, B = b | A = a_t, C = c_t)}{\sum_t P(B = b | A = a_t, C = c_t)}$$

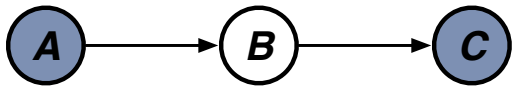
Simplify:

$$P(C = c | B = b) \longleftarrow \frac{\sum_t I(c, c_t) P(B = b | A = a_t, C = c_t)}{\sum_t P(B = b | A = a_t, C = c_t)}$$

Summary of EM algorithm

- **E-step** (Inference)

$$P(b|a_t, c_t) = \frac{P(c_t|b)\,P(b|a_t)}{\sum_{b'} P(c_t|b')\,P(b'|a_t)}$$



- **M-step** (Learning)

$$P(a) = \frac{1}{T}\,\mathrm{count}(A=a)$$

$$P(b|a) \longleftarrow \frac{\sum_t I(a, a_t)\,P(b|a_t, c_t)}{\sum_t I(a, a_t)}$$

$$P(c|b) \longleftarrow \frac{\sum_t I(c, c_t)\,P(b|a_t, c_t)}{\sum_t P(b|a_t, c_t)}$$

- **Convergence**

  There are no learning rates to tune.
  Each update increases the incomplete data log-likelihood:

$$\mathcal{L} = \sum_t \log \sum_b P(a_t)\,P(b|a_t)\,P(c_t|b)$$

Check In

Q. How much of EM did you understand?

A. (Nearly) All of it

B. Some of it, but I have some doubts

C. Maybe a little, but I'm pretty confused

D. Almost none of it; I'm totally lost

That’s all folks!